

1. To help you practice strategies for machine learning, this week we'll present another scenario and ask how you would act. We think this "simulator" of working in a machine learning project will give an idea of what leading a machine learning project could be like!

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You are employed by a startup building self-driving cars. You are in charge of detecting road signs (stop sign, pedestrian crossing sign, construction ahead sign) and traffic signals (red and green lights) in images. The goal is to recognize which of these objects appear in each image. As an example, this image contains a pedestrian crossing sign and red traffic lights.



$$y^{(i)} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \begin{matrix} \text{"stop sign"} \\ \text{"pedestrian crossing sign"} \\ \text{"construction ahead sign"} \\ \text{"red traffic light"} \\ \text{"green traffic light"} \end{matrix}$$

Your 100,000 labeled images are taken using the front-facing camera of your car. This is also the distribution of data you care most about doing well on. You think you might be able to get a much larger dataset off the internet, which could be helpful for training even if the distribution of internet data is not the same.

You are getting started with this project.

What is the first thing you do?

Assume each of the steps below would take about an equal amount of time (a few days).

- ☐ Spend a few days collecting more data using the front-facing camera of your car, to better understand how much data per unit time you can collect.
- ☐ Spend some time searching the internet for the data most similar to the conditions you expect on production.
- ☐ Invest a few days in thinking on potential difficulties, and then some more days brainstorming about possible solutions, before training any model.
- ☒ Train a basic model and do error analysis.

2. Your goal is to detect road signs (stop sign, pedestrian crossing sign, construction ahead sign) and traffic signals (red and green lights) in images. The goal is to recognize which of these objects appear in each image. You plan to use a deep neural network with ReLU units in the hidden layers.

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For the output layer, which of the following gives you the most appropriate activation function?

- ☐ ReLU
- ☐ Softmax
- ☐ Linear
- ☒ Sigmoid

3. True or False: When trying to determine what strategy to implement to improve the performance of a model, you manually check all images of the training set where the algorithm was successful.

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- ☐ True
- ☒ False

4. After working on the data for several weeks, your team ends up with the following data:

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- 100,000 labeled images taken using the front-facing camera of your car.
- 900,000 labeled images of roads downloaded from the internet.
- Each image's labels precisely indicate the presence of any specific road signs and traffic signals or

combinations of them. For example, $y^{(i)} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ means the image contains a stop sign and a red traffic light.

True or False: Because this is a multi-task learning problem, when an image is not fully labeled (for

example: $\begin{pmatrix} 0 \\ ? \\ ? \\ 1 \\ 0 \end{pmatrix}$) you can use it if you ignore those entries when calculating the loss function.

- ☐ False
- ☒ True

5. The distribution of data you care about contains images from your car's front-facing camera; which comes from a different distribution than the images you were able to find and download off the internet.

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How should you split the dataset into train/dev/test sets?

- ☐ Choose the training set to be the 900,000 images from the internet along with 20,000 images from your car's front-facing camera. The 80,000 remaining images will be split equally in dev and test sets.
- ☒ Choose the training set to be the 900,000 images from the internet along with 80,000 images from your car's front-facing camera. The 20,000 remaining images will be split equally in dev and test sets.
- ☐ Mix all the 100,000 images with the 900,000 images you found online. Shuffle everything. Split the 1,000,000 images dataset into 600,000 for the training set, 200,000 for the dev set and 200,000 for the test set.
- ☐ Mix all the 100,000 images with the 900,000 images you found online. Shuffle everything. Split the 1,000,000 images dataset into 980,000 for the training set, 10,000 for the dev set and 10,000 for the test set.

6. Assume you've finally chosen the following split between the data:

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Dataset:	Contains:	Error of the algorithm:
Training	940,000 images randomly picked from (900,000 internet images + 60,000 car's front-facing camera images)	12%
Training-Dev	20,000 images randomly picked from (900,000 internet images + 60,000 car's front-facing camera images)	15.1%
Dev	20,000 images from your car's front-facing camera	12.6%
Test	20,000 images from the car's front-facing camera	15.8%

You also know that human-level error on the road sign and traffic signals classification task is around 0.5%. Which of the following is true?

- ☐ You have a large data-mismatch problem.
- ☒ You have a high bias.
- ☐ You have a high variance problem.
- ☐ You have a too low avoidable bias.

7. Assume you've finally chosen the following split between the data:

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Dataset:	Contains:	Error of the algorithm:
Training	940,000 images randomly picked from (900,000 internet images + 60,000 car's front-facing camera images)	2%
Training-Dev	20,000 images randomly picked from (900,000 internet images + 60,000 car's front-facing camera images)	2.3%
Dev	20,000 images from your car's front-facing camera	1.3%

Test	20,000 images from the car's front-facing camera	1.1%
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Human-level error on this task is approximately 0.5%. (Bayes error is the lowest possible error rate for a task. Human-level error is a good estimation of Bayes error.)

True or False: Based on this, the Bayes error for the car camera images (Dev/Test) is higher than the Bayes error for the mixed internet/car images (Training).

- ☒ False
☐ True

8. You decide to focus on the dev set and check by hand what the errors are due to. Here is a table summarizing your discoveries:

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Overall dev set error	15.3%
Errors due to incorrectly labeled data	4.1%
Errors due to foggy pictures	3.0%
Errors due to partially occluded elements.	7.2%
Errors due to other causes	1.0%

In this table, 4.1%, 7.2%, etc. are a fraction of the total dev set (not just examples of your algorithm mislabeled). For example, about $7.2/15.3 \approx 47\%$ of your errors are due to partially occluded elements.

True or False: You shouldn't invest all your efforts to get more images with partially occluded elements since $4.1 + 3.0 + 1.0 = 8.1 > 7.2$.

- ☒ False
☐ True

9. You can buy a specially designed windshield wiper that helps wipe off some of the raindrops on the front-facing camera.

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Overall dev set error	15.3%
Errors due to incorrectly labeled data	4.1%
Errors due to foggy pictures	8.0%
Errors due to rain drops stuck on your car's front-facing camera	2.2%
Errors due to other causes	1.0%

Which one of the following statements do you agree with?

- ☐ 2.2% would be a reasonable estimate of the minimum amount this windshield wiper could improve performance.
☒ 2.2% would be a reasonable estimate of the maximum amount this windshield wiper could improve performance.
☐ 2.2% would be a reasonable estimate of how much this windshield wiper will improve performance.
☐ 2.2% would be a reasonable estimate of how much this windshield wiper could worsen performance in the worst case.

10. You decide to use data augmentation to address foggy images. You find 1,000 pictures of fog off the internet, and "add" them to clean images to synthesize foggy days, like this:

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True or False: We can't use this data since they have a different distribution from the ones we used (internet and front-facing camera).

- ☐ True
☒ False

11. After working further on the problem, you've decided to correct the incorrectly labeled data on the dev set. Which of these statements do you agree with?

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- ☐ You should not correct the incorrectly labeled data in the test set, so that the dev and test sets continue to come from the same distribution.
☒ You should also correct the incorrectly labeled data in the test set, so that the dev and test sets continue to come from the same distribution.
☒ You should correct incorrectly labeled data in the training set as well to avoid your training set now being even more different from your dev set.
☐ You do not necessarily need to fix the incorrectly labeled data in the training set because it's acceptable for the training set distribution to differ from the dev and test sets. Note that it is important that the dev set and test set have the same distribution.

12. Your client asks you to add the capability to detect dogs that may be crossing the road to the system. He can provide a relatively small set containing dogs. Which of the following do you agree most with?

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- ☐ You can use weights pre-trained on the original data, and fine-tune with the data now including the dogs.
☐ Using pre-trained weights can severely hinder the ability of the model to detect dogs since they have too many learned features.
☐ You will have to re-train the whole model now including the dogs' data.
☒ You should train a single new model for the dogs' task, and leave the previous model as it is.

13. One of your colleagues at the startup is starting a project to classify road signs as stop, dangerous curve, construction ahead, dead end, and speed limit signs. He has approximately 30,000 examples of each image and 30,000 images without a sign.

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True or False: This case could benefit from using multi-task learning.

- ☐ False
☒ True

14. You're building a system to recognize stop signs. Your approach is:

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First, a neural network predicts bounding box coordinates around potential traffic signs in an image. Second, a separate neural network determines if each predicted sign is a stop sign.

Is this an example of multi-task learning?

- ☐ True
☒ False

15. **True or False:** An end-to-end approach doesn't require that we hand-design useful features; it only requires a large enough model.

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- ☐ False
☒ True